



OPTICAL COMMUNICATIONS

Pierpaolo Boffi



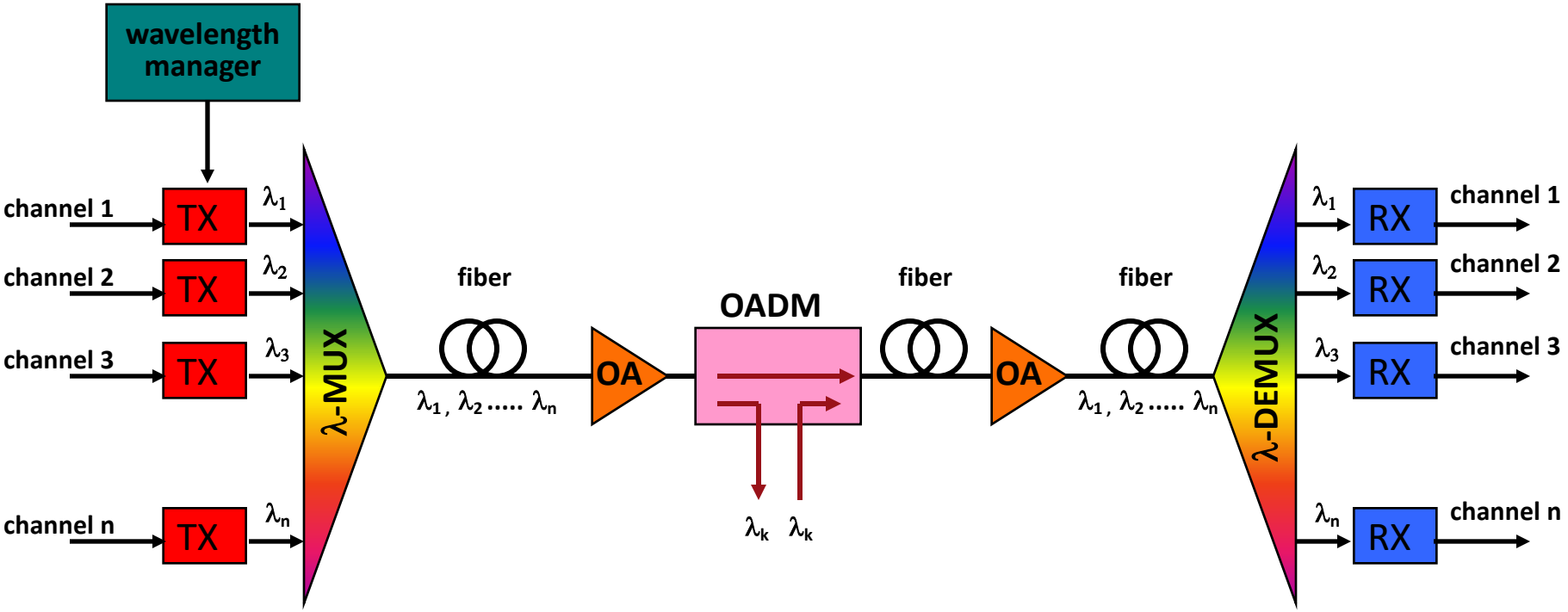
POLITECNICO
MILANO 1863



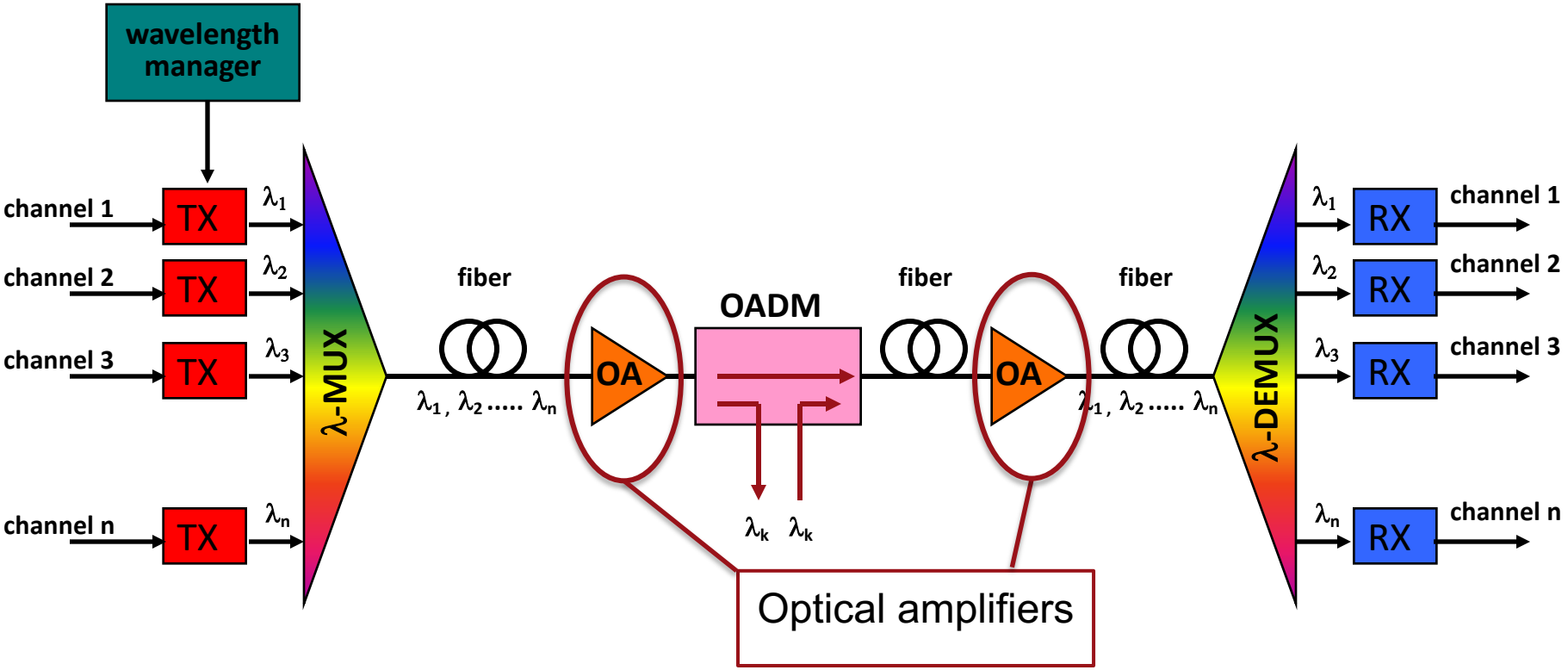
9 – Optical Amplification

Optical amplification basics, EDFA, gain and saturation, ASE and NF, Raman amplification

WDM system

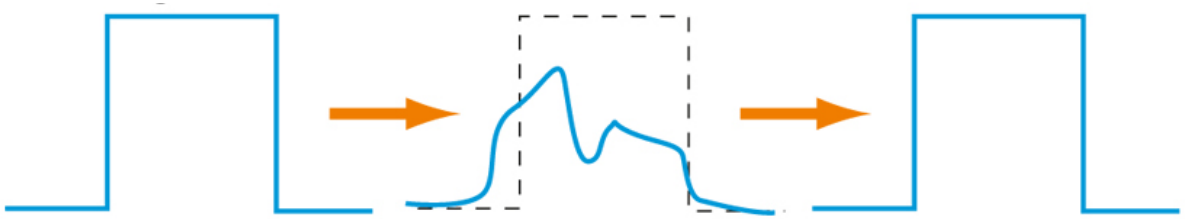

wavelength manager
channel 1
channel 2
channel 3
channel n
TX
TX
TX
TX
λ_1
λ_2
λ_3
λ_n
λ-MUX
fiber
$\lambda_1, \lambda_2 ..... \lambda_n$
OA
OADM
λ_k λ_k
fiber
OA
fiber
$\lambda_1, \lambda_2 ..... \lambda_n$
λ-DEMUX
λ_1
λ_2
λ_3
λ_n
RX
RX
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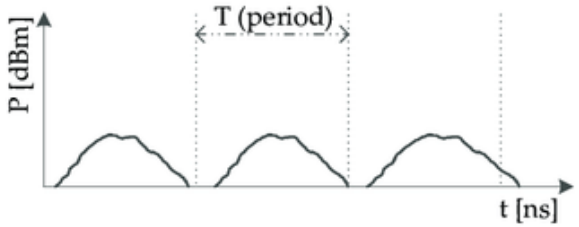
WDM system


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λ_3
λ_n
RX
RX
RX
RX
channel 1
channel 2
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Optical amplifiers

Signal REGENERATION

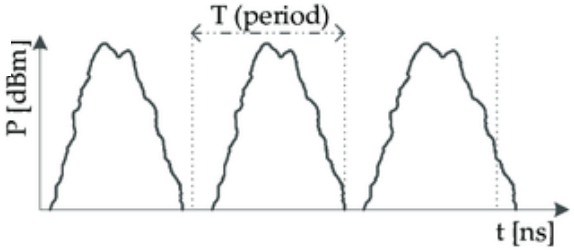
REGENERATION

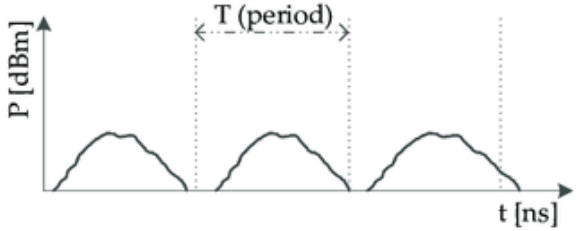




AMPLIFICATION

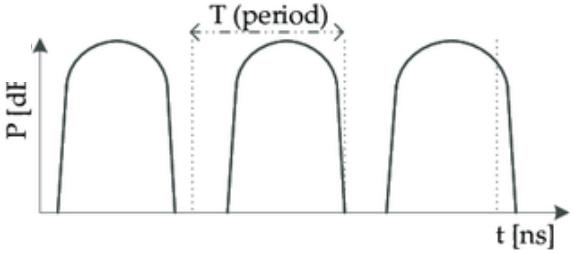


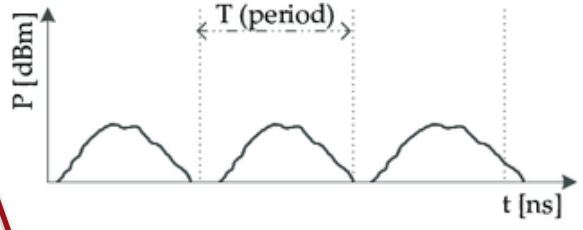




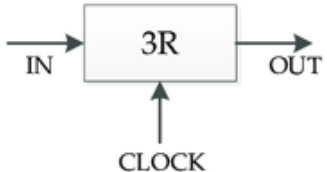
RESHAPING

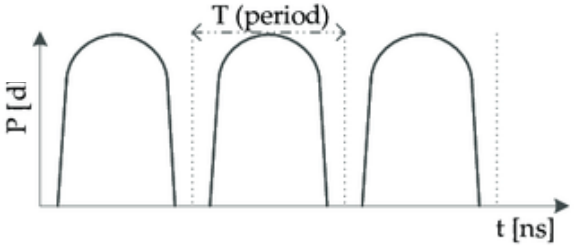




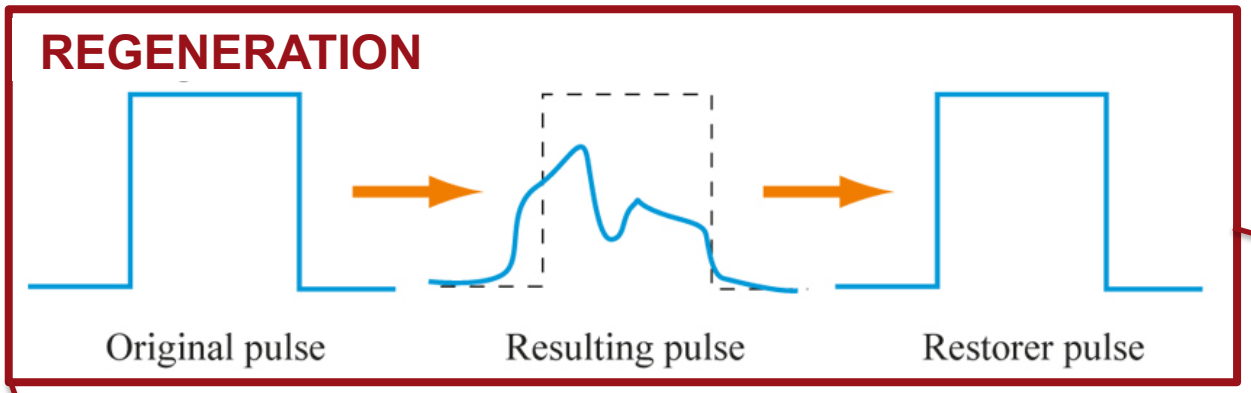


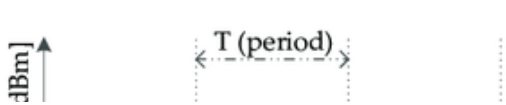
RETIMING



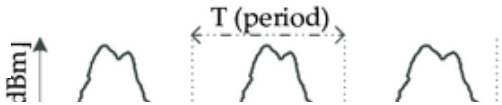


Signal REGENERATION

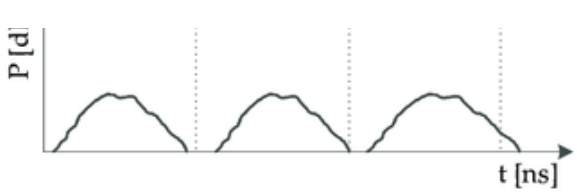




AMPLIFICATION

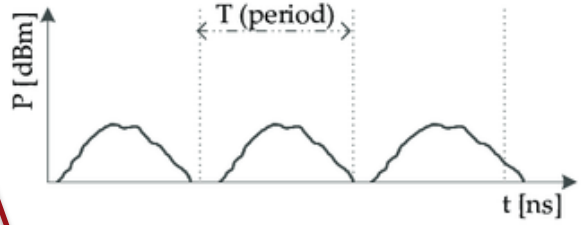


It is necessary to receive the optical signal, converting it from an optical pulse to an electrical one, then after the reshaping and the retiming, to convert again from electrical to optical signal.

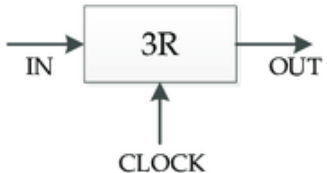


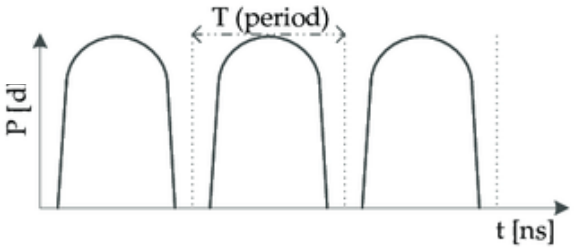






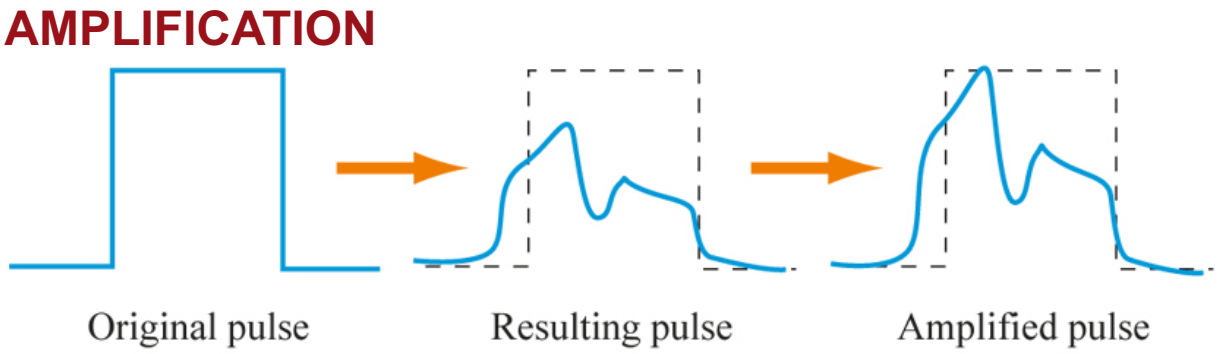
RETIMING



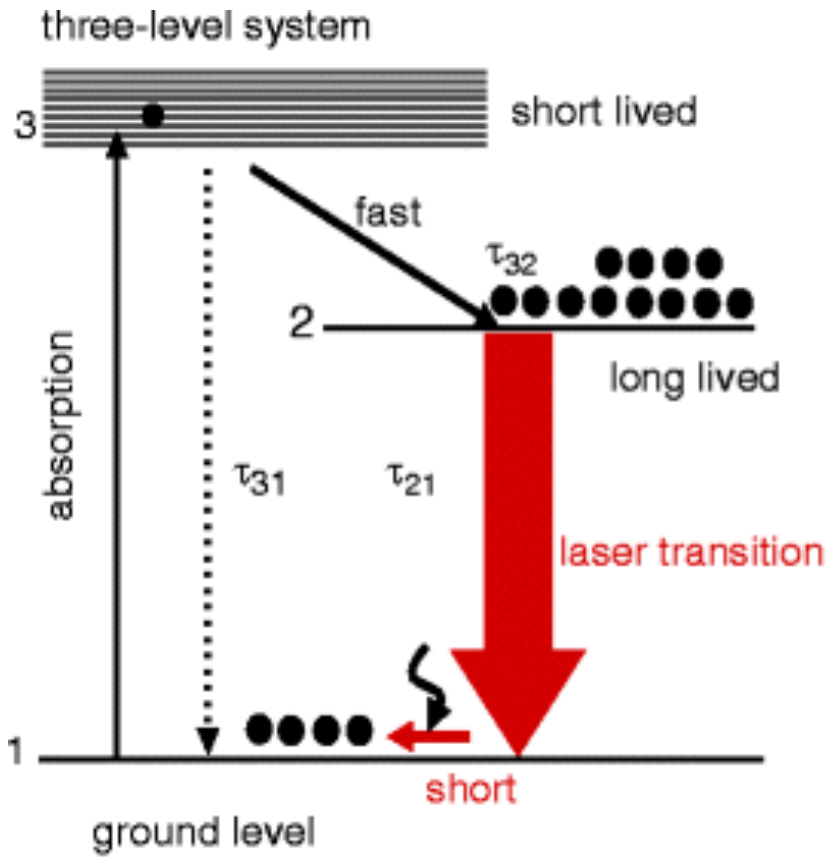


Signal AMPLIFICATION

The only AMPLIFICATION can be realized in an all-optical transparent way, without any optical-to-electrical and electrical-to-optical convertion.

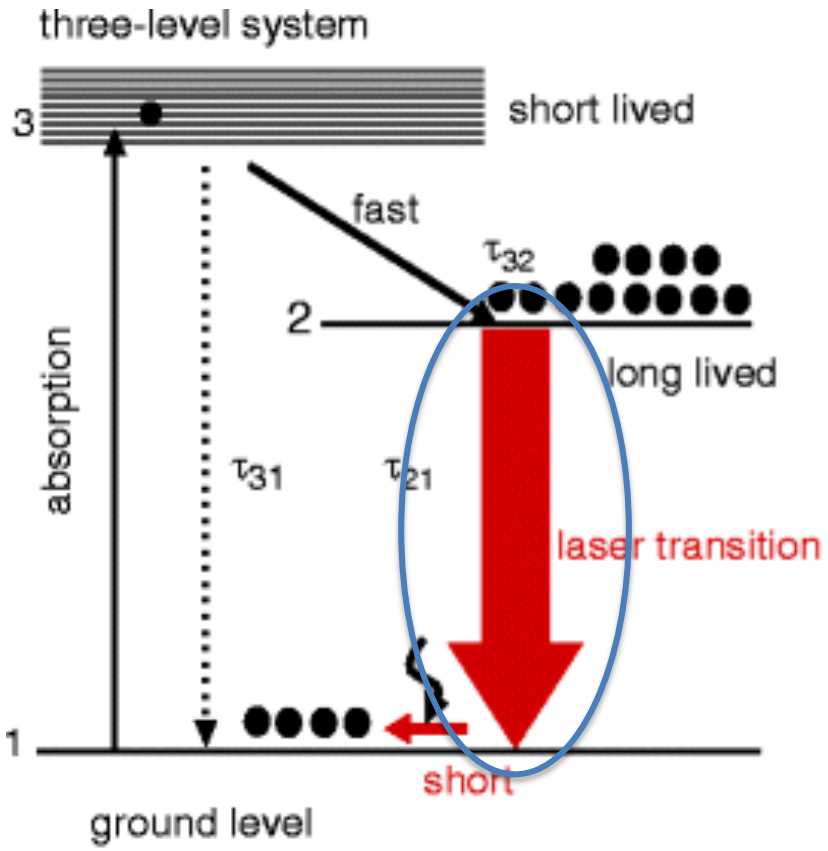


Optical amplification I



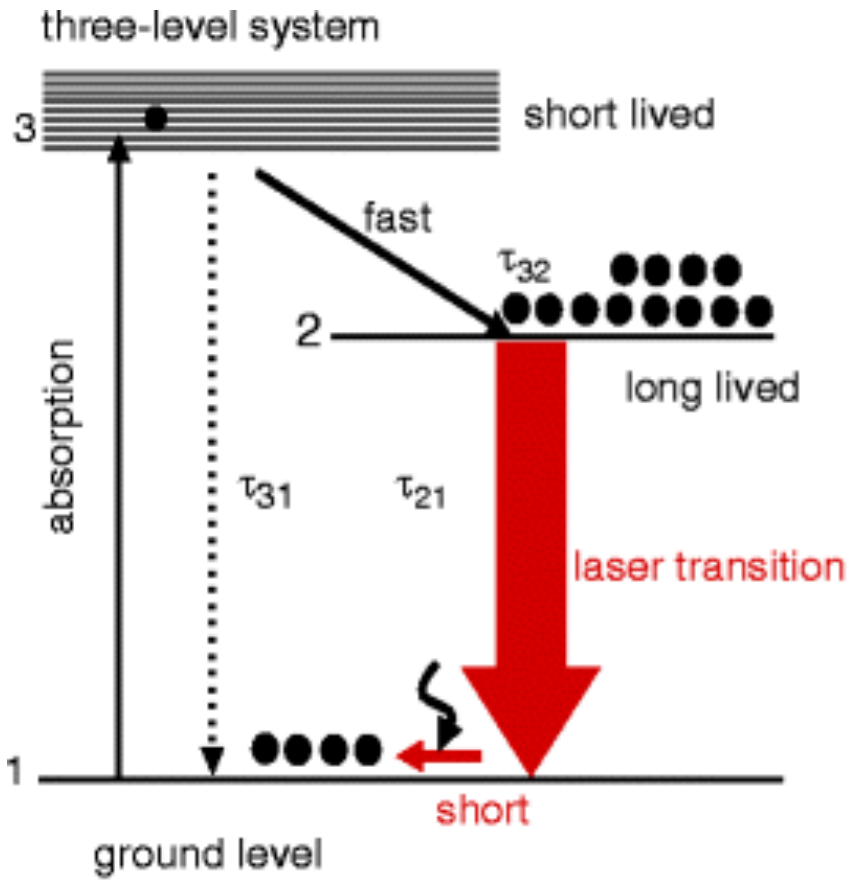
- The **electron absorbs the energy from the photon** and **is excited** to an unstable energy state (3).
- From this level, the **electrons decay to the lower state rapidly**: the energy emitted in this decay is absorbed in the form of molecular vibrations.
- The electrons remain in the new **intermediate (metastable) energy state** for a relatively long time in atomic terms.

Optical amplification II



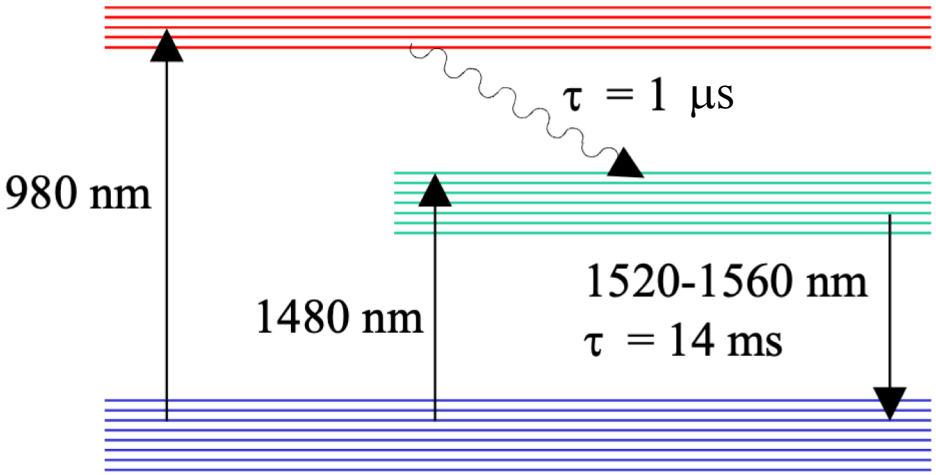
- The **decay can be stimulated by an incident photon.**
- The emitted photon has exactly the **same energy (λ), phase and direction as the photon** that caused its emission (**stimulated emission**).
- This is the basis of the **amplification process.**

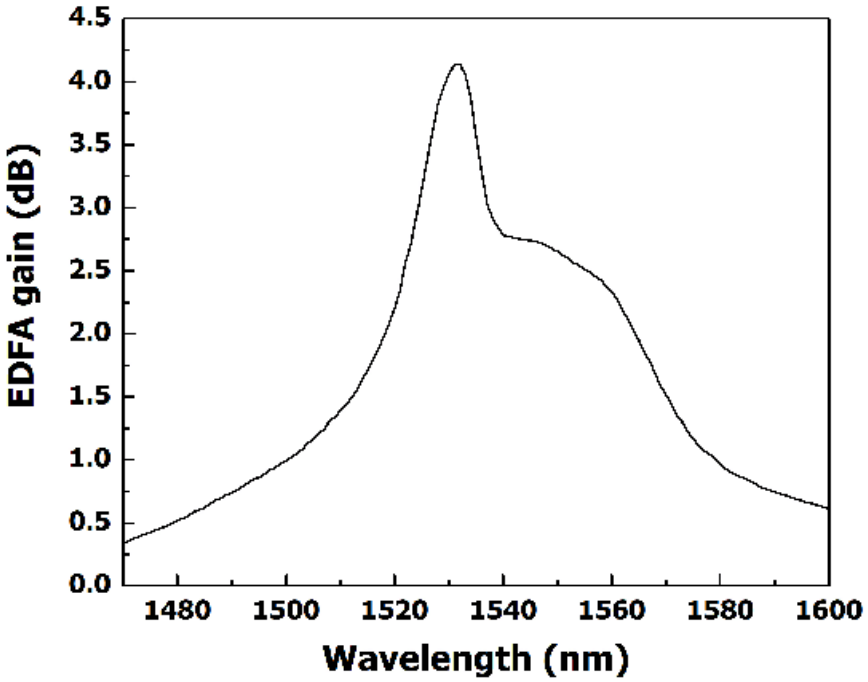
Optical amplification III



- An electron can spontaneously decay from the metastable state by emitting a photon at the band gap (**spontaneous emission**), even in the absence of an input photon.
- This produces an unavoidable optical noise called **Amplified Spontaneous Emission (ASE).**

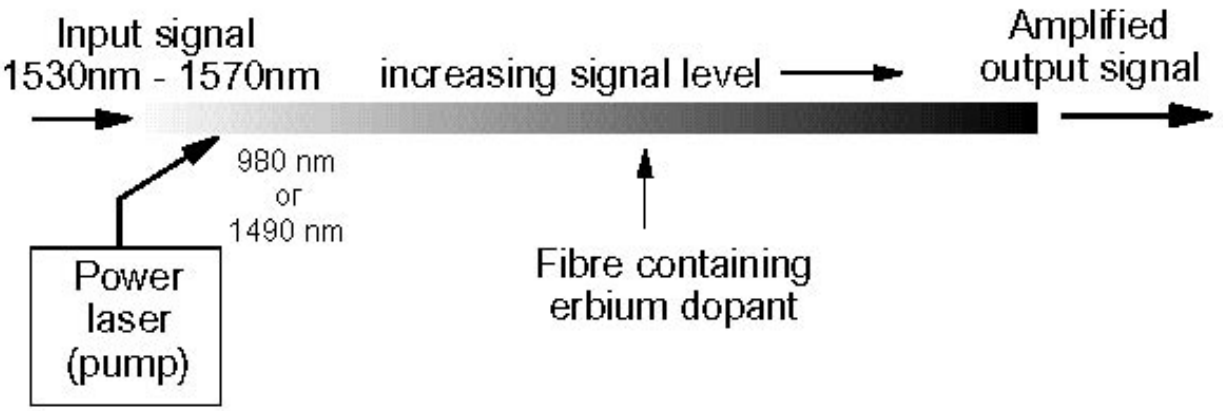
Erbium-Doped Fiber Amplifier – EDFA





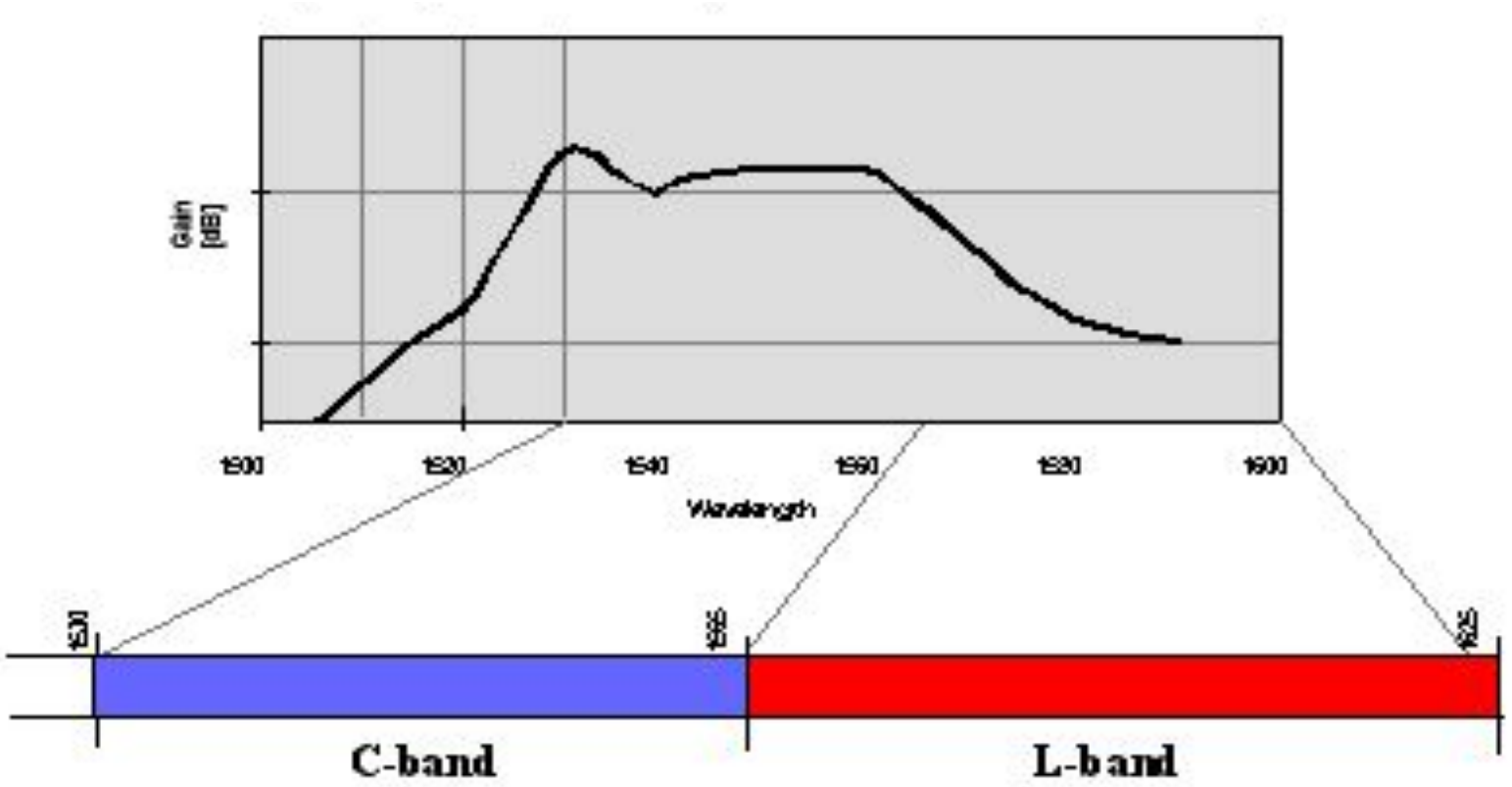
- Possibility to amplify third-window signals by using a fiber doped with the chemical element **Erbium (Er)**, which has an electronic transition exactly coinciding with the average energy possessed by third-window photons.
- When dispersed in a glass matrix, the amplifying transition corresponds to a **quasi-continuous band from 1525 nm to 1570 nm (C-band: 1530 – 1565 nm).**

Erbium-Doped Fiber Amplifier – EDFA



- The Erbium-doped fiber works as an optical amplifier for the signals travelling inside the doped fiber itself.
- **Optical pumping** with radiation at **980 nm** (more efficient) or at **1480 nm**

EDFA amplification spectrum

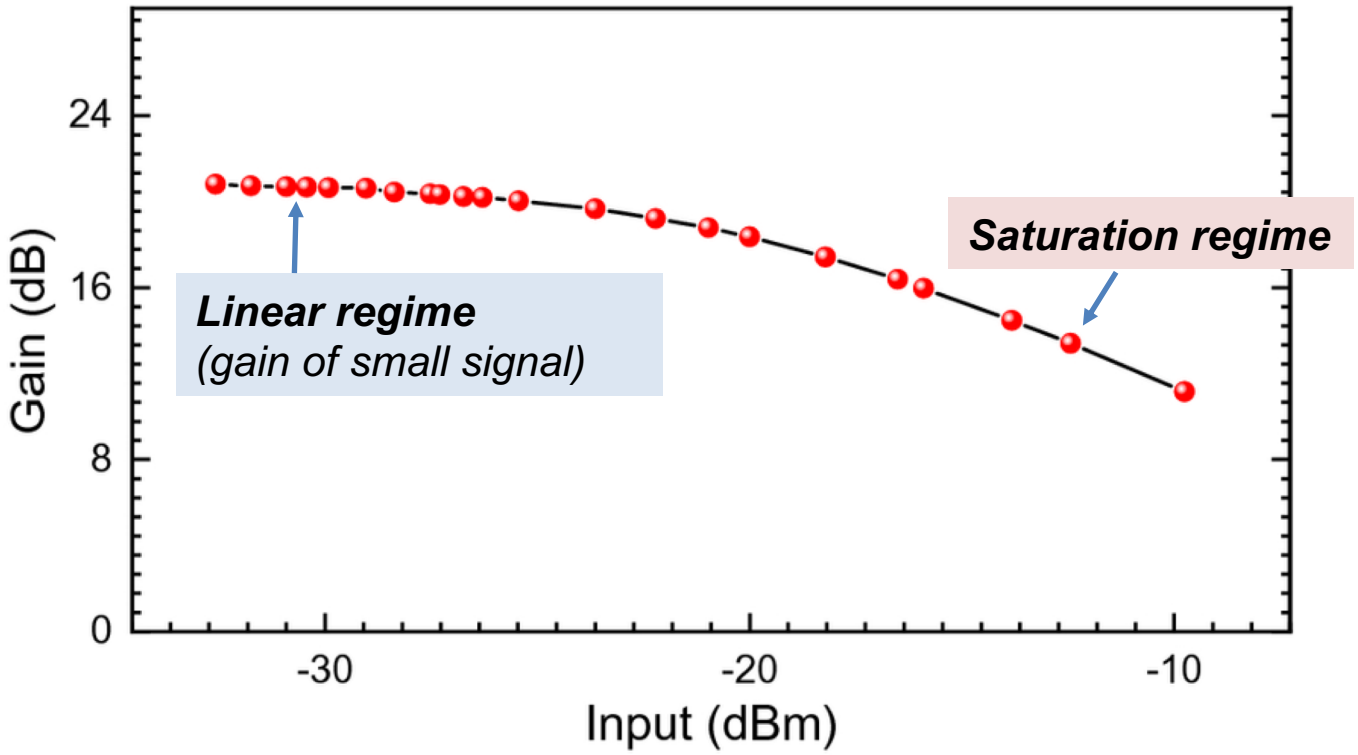


C-band (conventional): 1530 – 1565 nm
L- band (long): 1565 – 1625 nm

GAIN

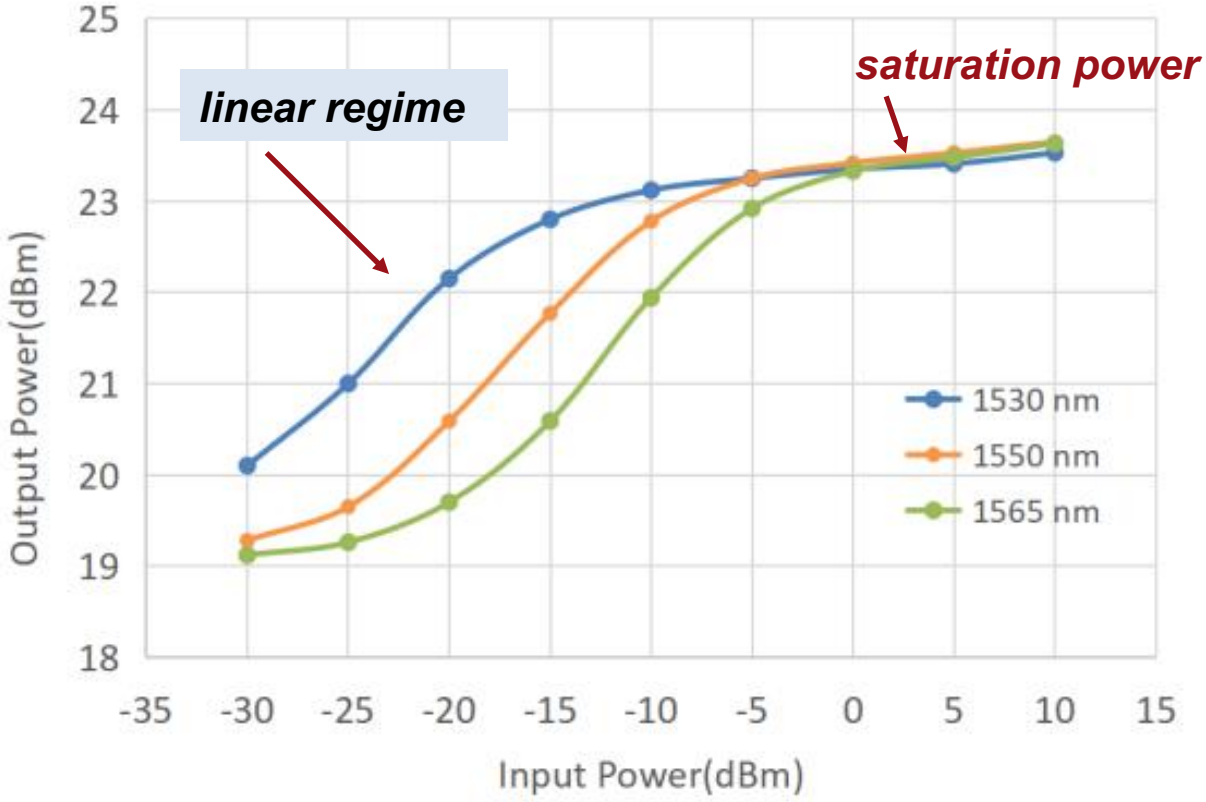
➢ **GAIN**: ratio between the input power and the output power

$$G = \frac{P_{out}}{P_{in}} \rightarrow . G_{dB} = 10 \log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$



SATURATION

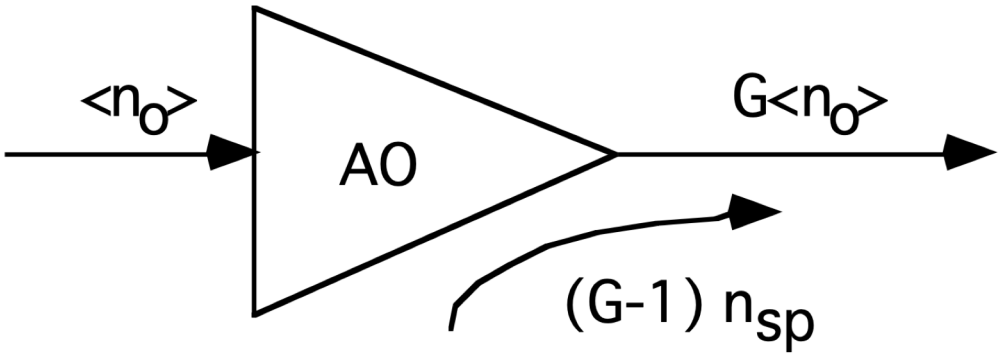
- **SATURATION POWER**: maximum output power (as the input power increases, the output power stops increasing). All the pump power is already used.



ASE noise

- **Amplified Spontaneous Emission - ASE:** the number of input photons is amplified with gain G while simultaneously introducing ASE noise equal to

$$w = n_{sp}(G-1)$$



where n_{sp} is the **spontaneous inversion factor**

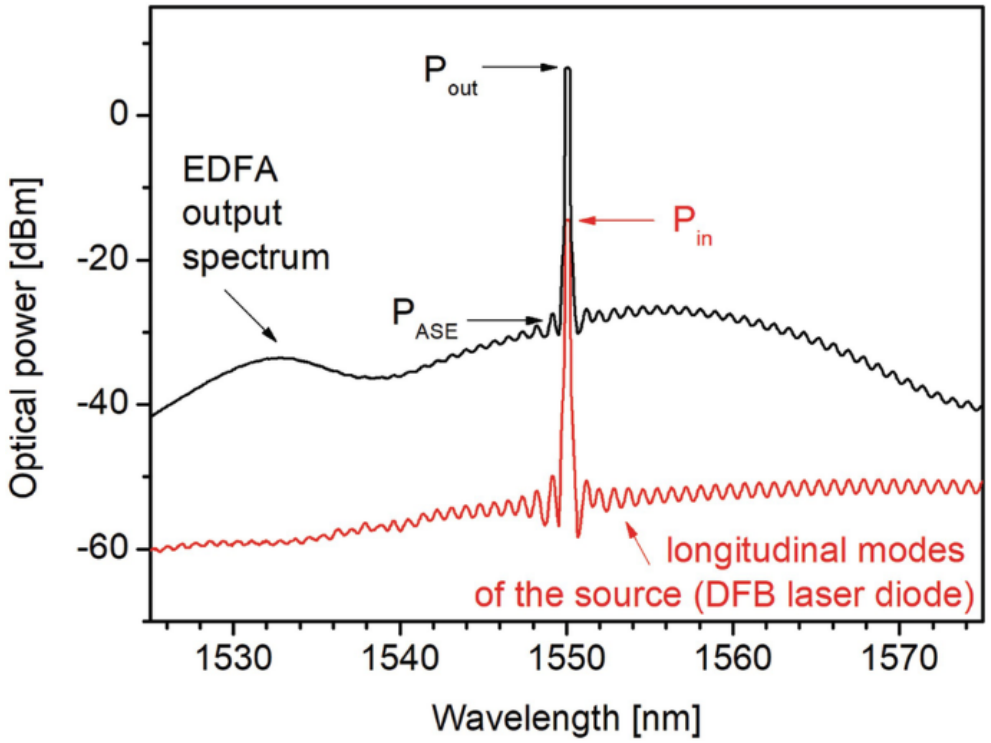
$$n_{sp} = \left(\frac{N_H}{N_H - N_L}\right) \xrightarrow{N_H \gg N_L} 1$$

ASE noise

- **Amplified Spontaneous Emission - ASE:** the number of input photons is amplified with gain G while simultaneously introducing ASE noise

$$P_{ASE} = 2h\upsilon \cdot \Delta\upsilon \cdot n_{sp}(G-1)$$

n_{sp} spontaneous inversion factor
h Plank's constant
υ optical frequency
$\Delta\upsilon$ optical band
G EDFA gain



Noise Figure - NF

➢ **Noise Figure - NF:** ratio between the SNR in input and the SNR in output

$$NF_{lin} = \frac{SNR_{in}}{SNR_{out}}$$

$$NF_{lin} = 2\eta \cdot n_{sp} \frac{(G-1)}{G}$$

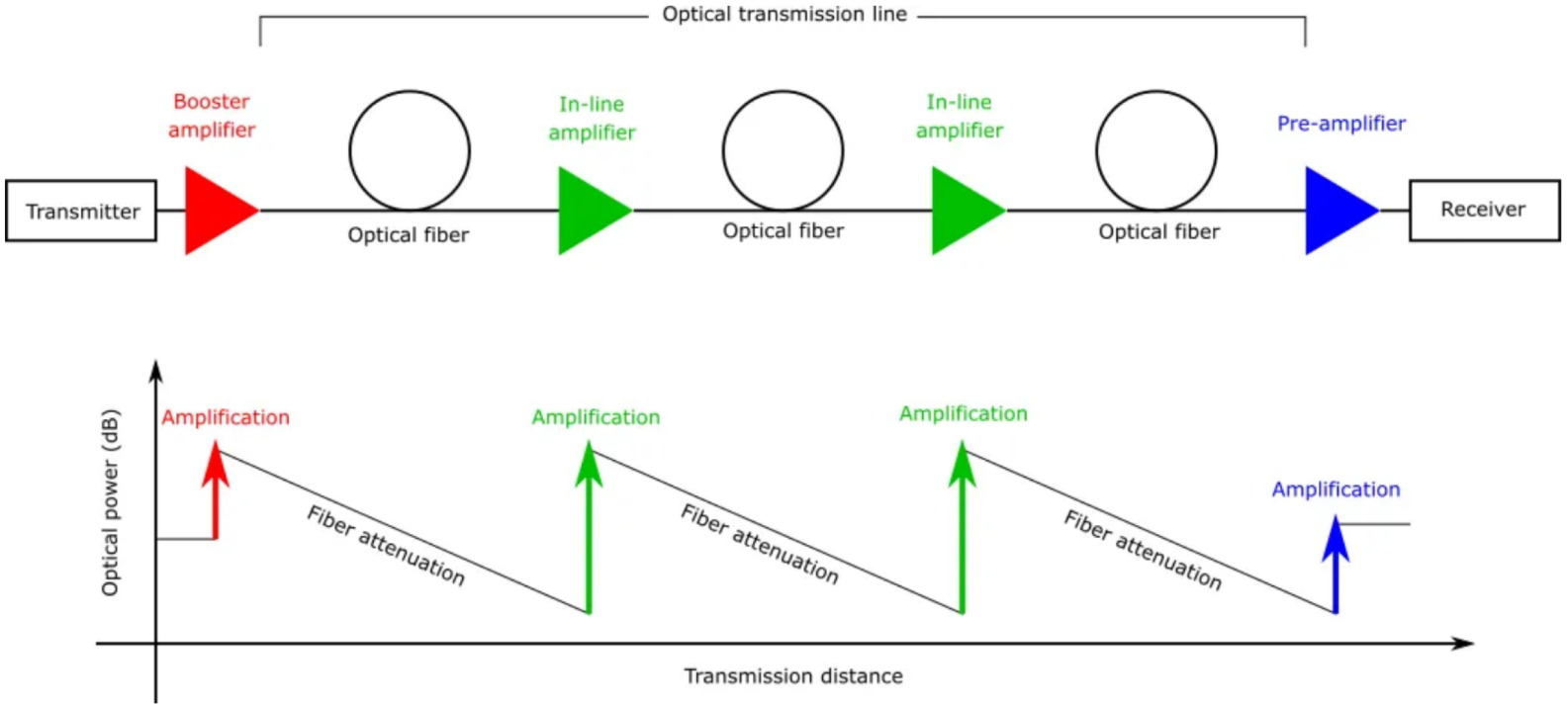
In case of ideal conversion photon-electron, efficiency $\eta = 1$

For G>>1 $NF_{lin} = 2n_{sp}$

When population inversion is full, $n_{sp} = 1$ ➡ $NF_{lin} = 2 = 3\ dB$

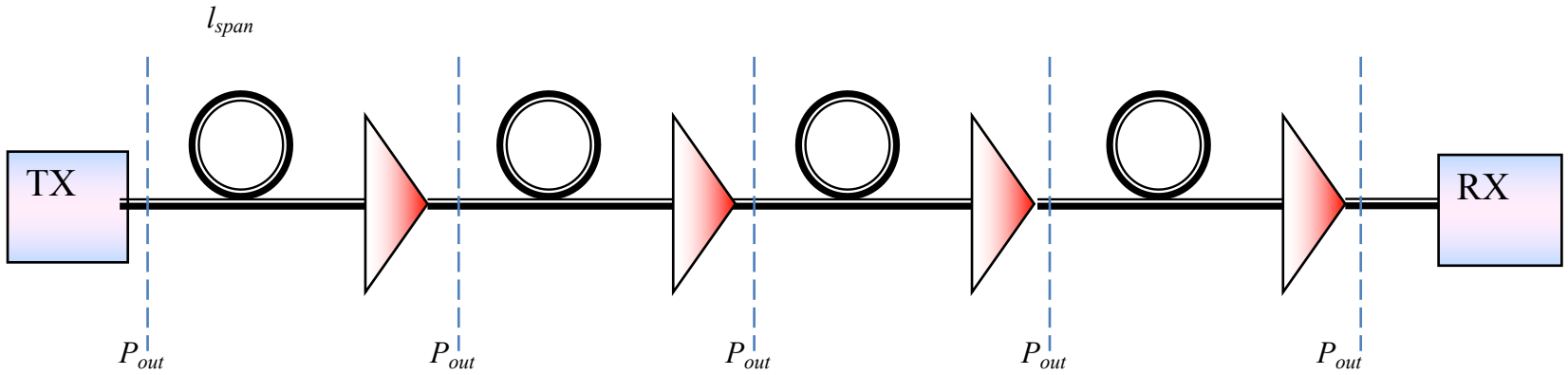
This is the ideal best value
Lowest limit NF

Optical amplifier in the communication system



- **Booster**: EDFA with high saturation power to inject high power in fiber input to compensate the propagation losses. High ASE ($NF \cong 5 - 6$), but attenuated by propagation.
- **Line amplifier**: EDFA between two spans to amplify the attenuated signal.
- **Pre-amplifier**: EDFA is necessary to face the electrical noise introduced by the receiver, but introducing ASE. They operate in small signal regime, with high G,requiring low NF ($NF \cong 3.5 - 4$).

Cascade of EDFAs in multi-span amplified system



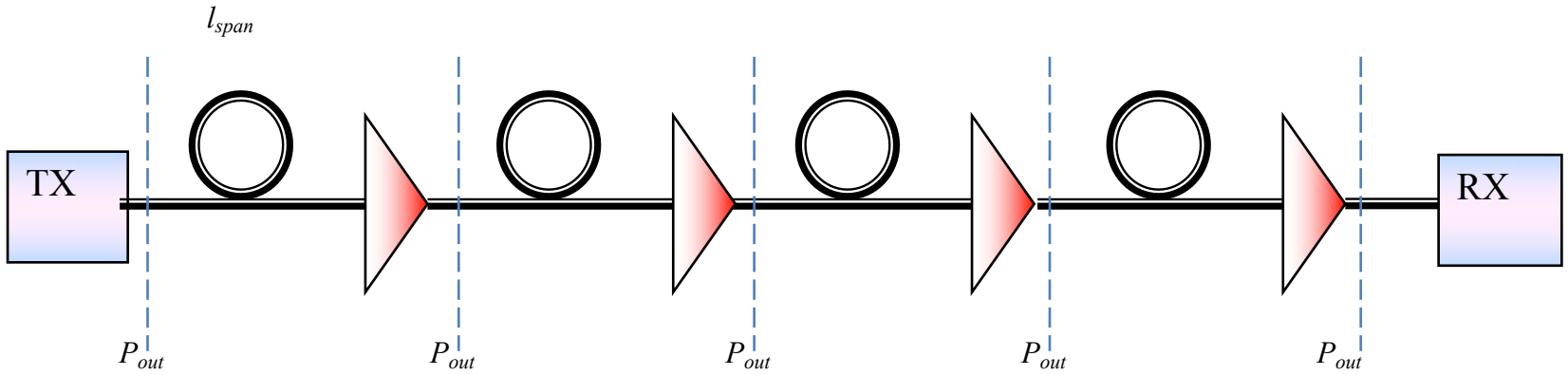
- Multi-span system with optical amplifiers: EDFAs are used to exactly compensate by the gain G the losses A due to α and the length of the single span l_{span} ➡ $A \cdot G = 1$
- The ASE noise contribution of each amplifier is added at the output and is amplified by the subsequent amplifiers.

$$P_{OUT_{EDFA(2)}} = A \cdot G \cdot P_{OUT_{EDFA(1)}} + A \cdot G \cdot P_{ASE_{(1)}} + P_{ASE_{(2)}} = P_{OUT} + P_{ASE_{(1)}} + P_{ASE_{(2)}}$$

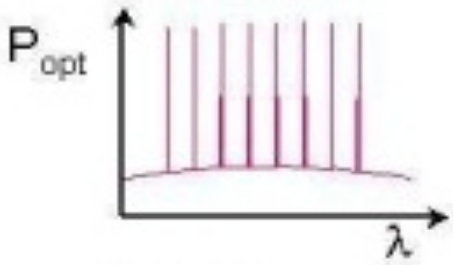
$$P_{OUT_{EDFA(3)}} = A \cdot G \cdot [P_{OUT} + P_{ASE_{(1)}} + P_{ASE_{(2)}}] + P_{ASE_{(3)}} = P_{OUT} + P_{ASE_{(1)}} + P_{ASE_{(2)}} + P_{ASE_{(3)}}$$

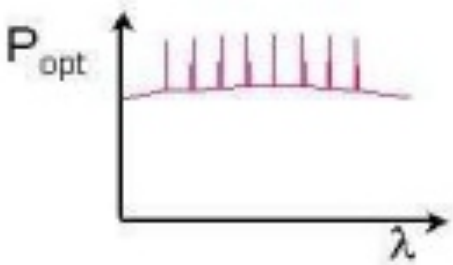
$$P_{OUT_{EDFA(n)}} = A \cdot G \cdot [P_{OUT} + \sum_1^{n-1} P_{ASE_{(j)}}] + P_{ASE_{(n)}} = P_{OUT} + \sum_1^{n} P_{ASE_{(j)}}$$

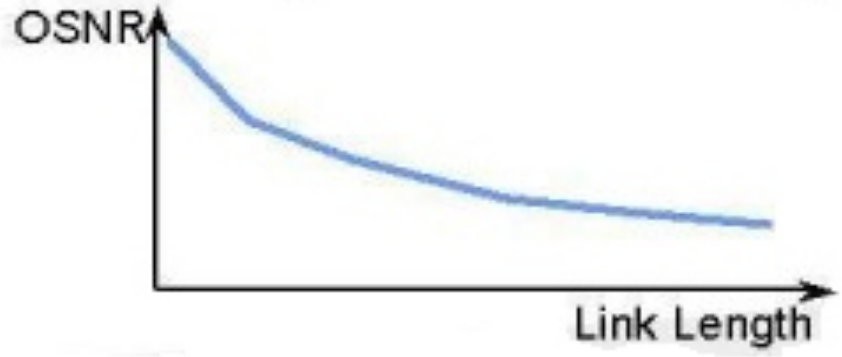
Cascade of EDFAs in multi-span amplified system



$$NF_{tot} = NF_1 + \frac{(NF_2 - 1)}{G_1} + \cdots \frac{(NF_n - 1)}{G_1 G_2 \cdots G_{n-1}}$$

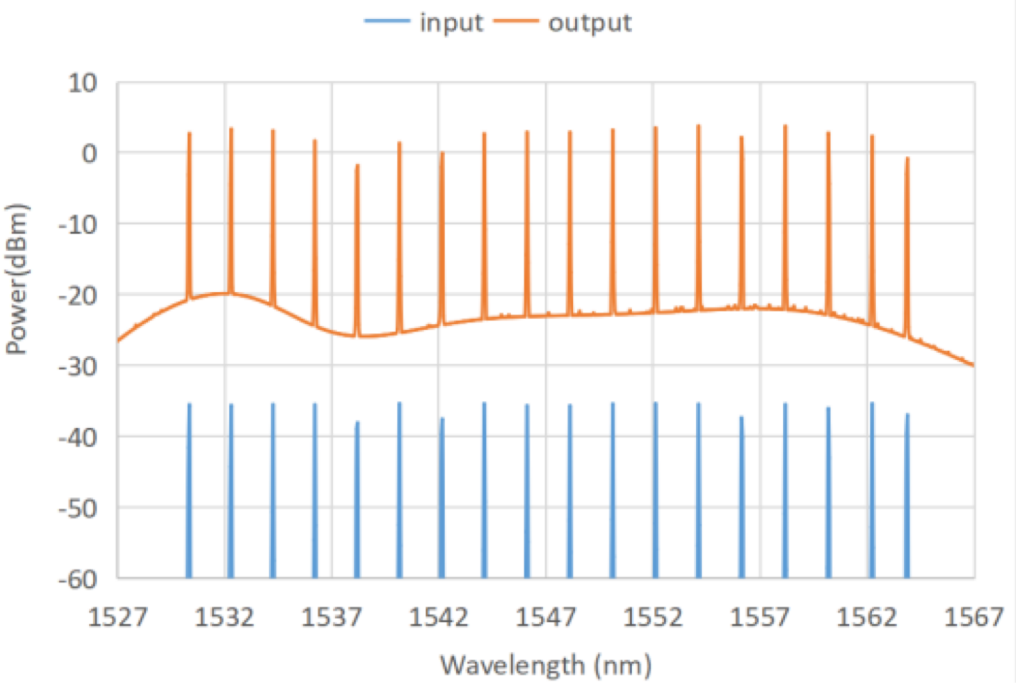






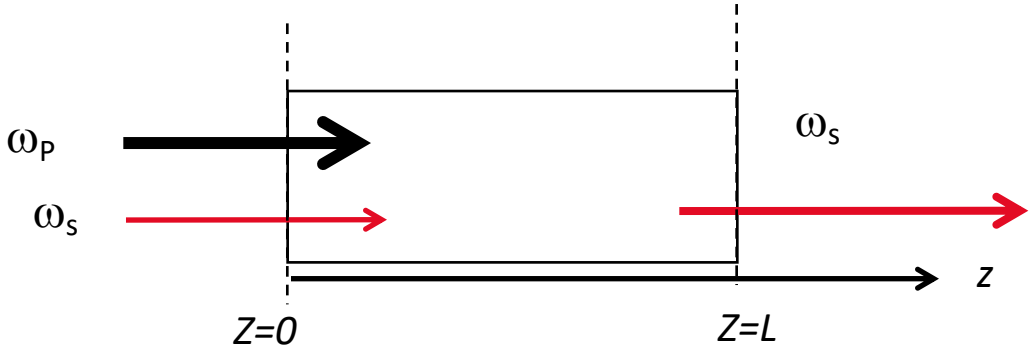
EDFA amplification in WDM transmission

- **WDM system**: simultaneous amplification of all the WDM channels. If the amplifier works in saturation, the saturation power output from the amplifier is divided among all channels.
- If the saturation power of the optical amplifiers is $X\ dBm$ and the system has *2N* channels, each channel will have at the output of each amplifier a power equal to: $X\ dBm - 3\ N\ dB$.
- In the WDM case, an amplifier with a **very flat gain curve in λ** is required so as not to introduce a variable gain between channels.
- For common EDFAs, the amplification band is about **4 THz** (C-band, conventional band). The EDFA can be pushed to work even in unconventional areas (L-band, long band): 7 THz band.



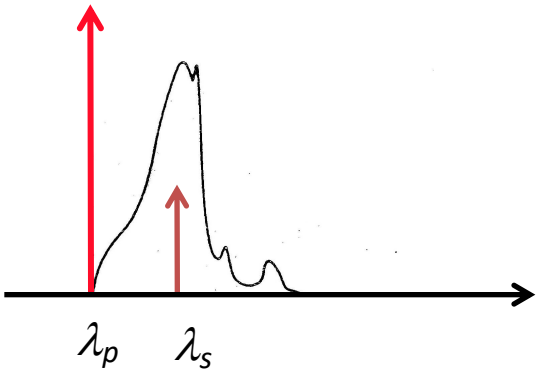
Optical amplification based on Raman effect

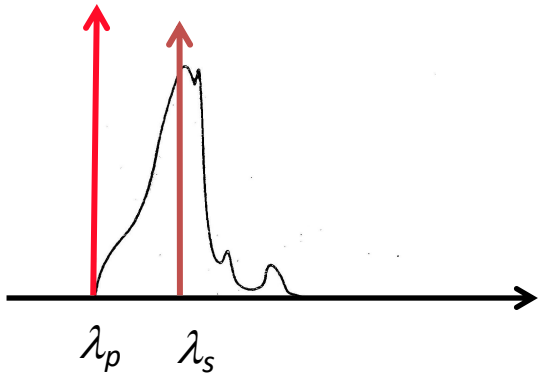
- **Stimulated Raman Scattering (SRS)** determines the generation of a new optical signal translated in frequency of 13.2 THz with respect to the pump.
- SRS can also determine the **amplification of a signal in the case the signal is at a frequency lower than about 13.2 THz with respect to the pump**. The amplification is more efficient the closer the frequency spacing is to 13.2 THz.
- In standard SMF this effect is small and becomes significant only for high pump powers and long propagated fiber link.

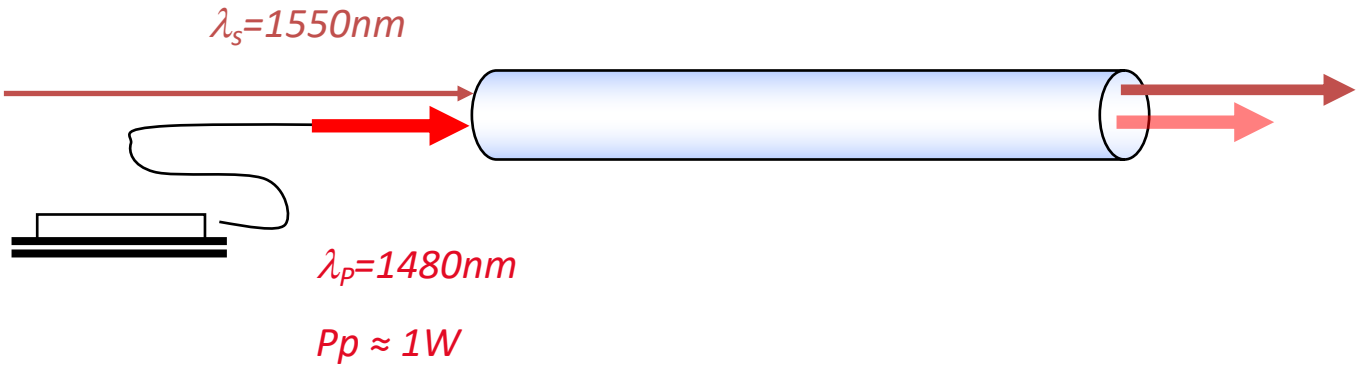


Raman amplification

- The **optical fiber itself can be used to amplify an optical signal** (no special doped fibers are necessary), if the signal is propagated together with the optical pump and the frequency of the signal falls within the Raman gain spectrum of the pump.

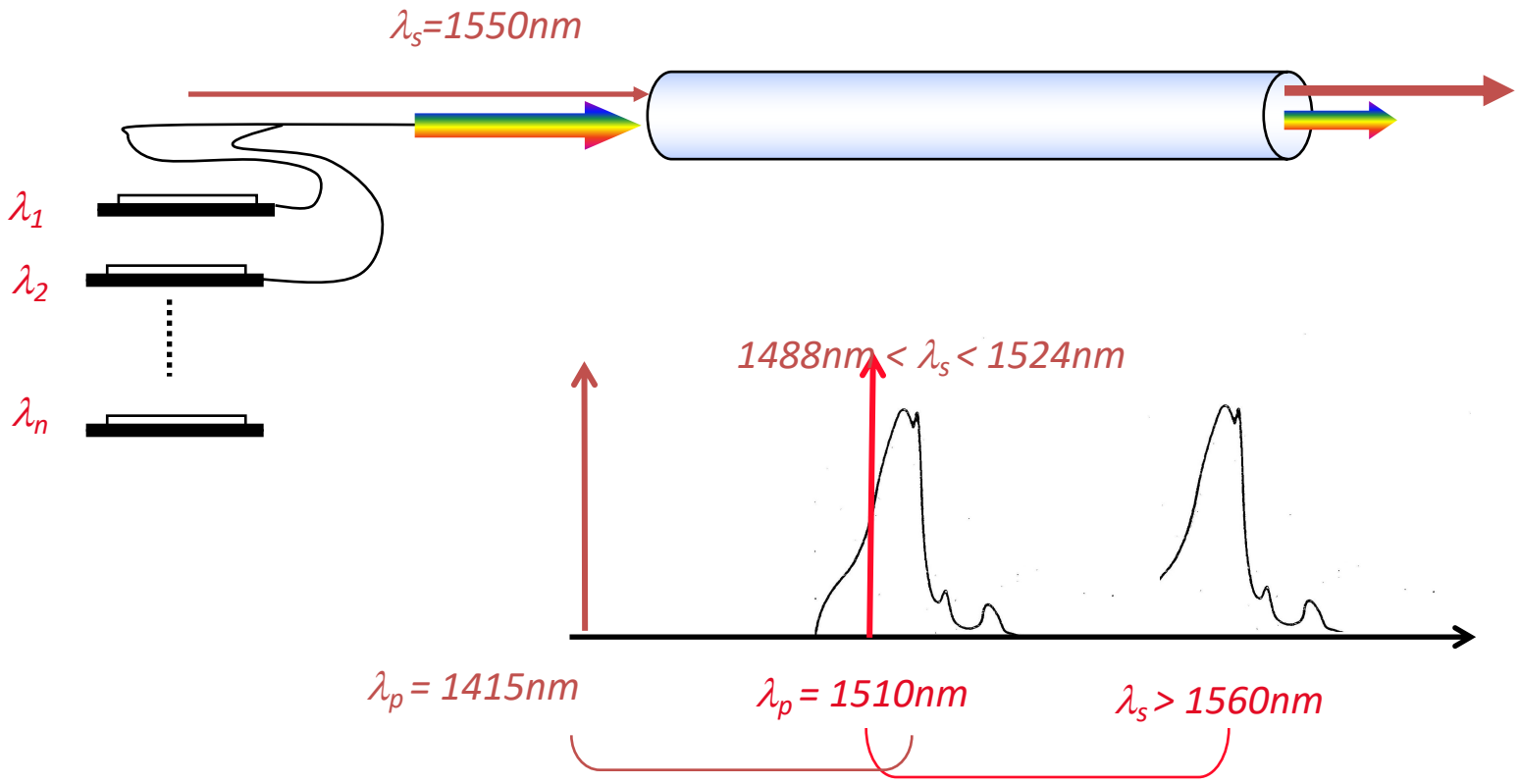






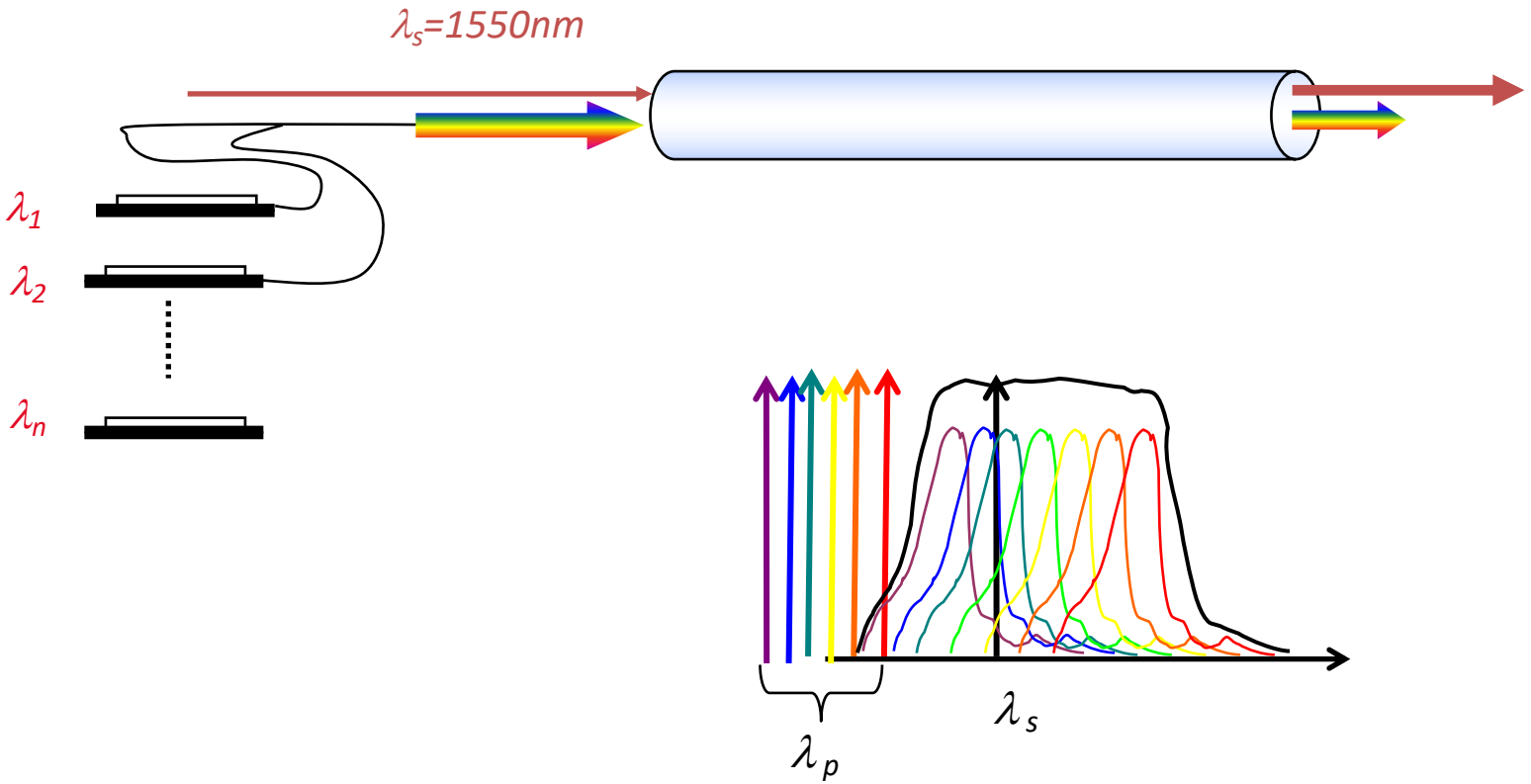
Raman amplification over multiple bands

- By injecting multiple pumps simultaneously at appropriate wavelengths, an amplifier can be realized operating **in multiple bands**, amplifying also the S and L bands.

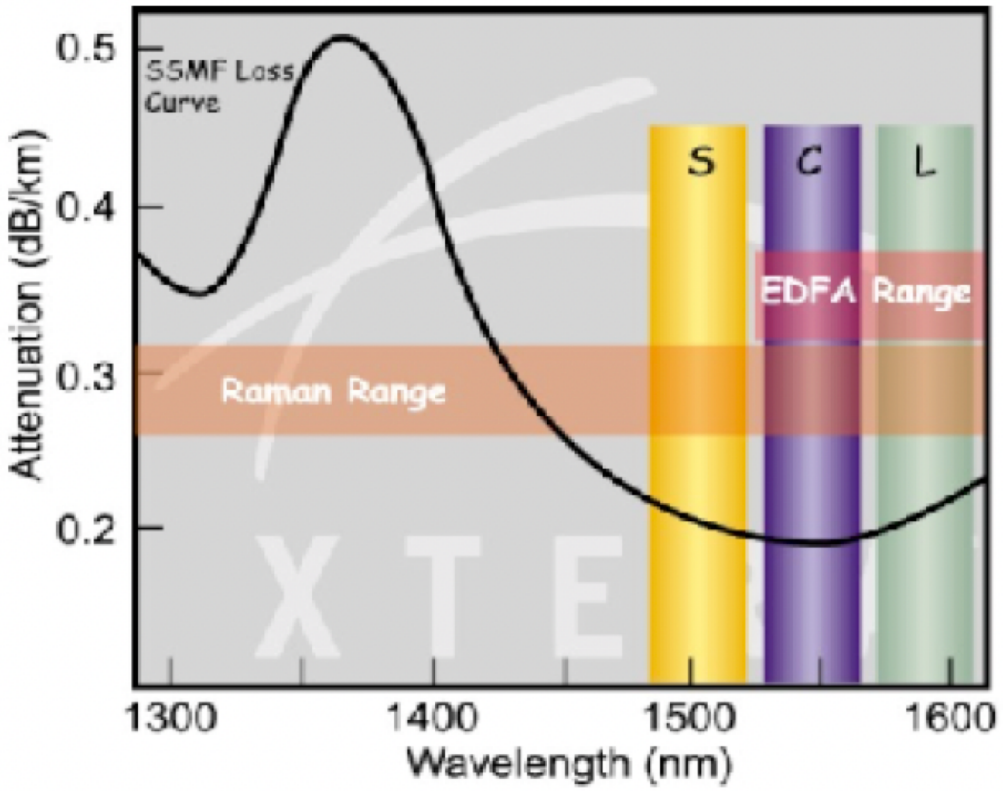


Raman amplification with broad gain spectrum

- By injecting multiple pumps at the same time and close enough in wavelength, it is possible to generate a **very broad amplification spectrum**.

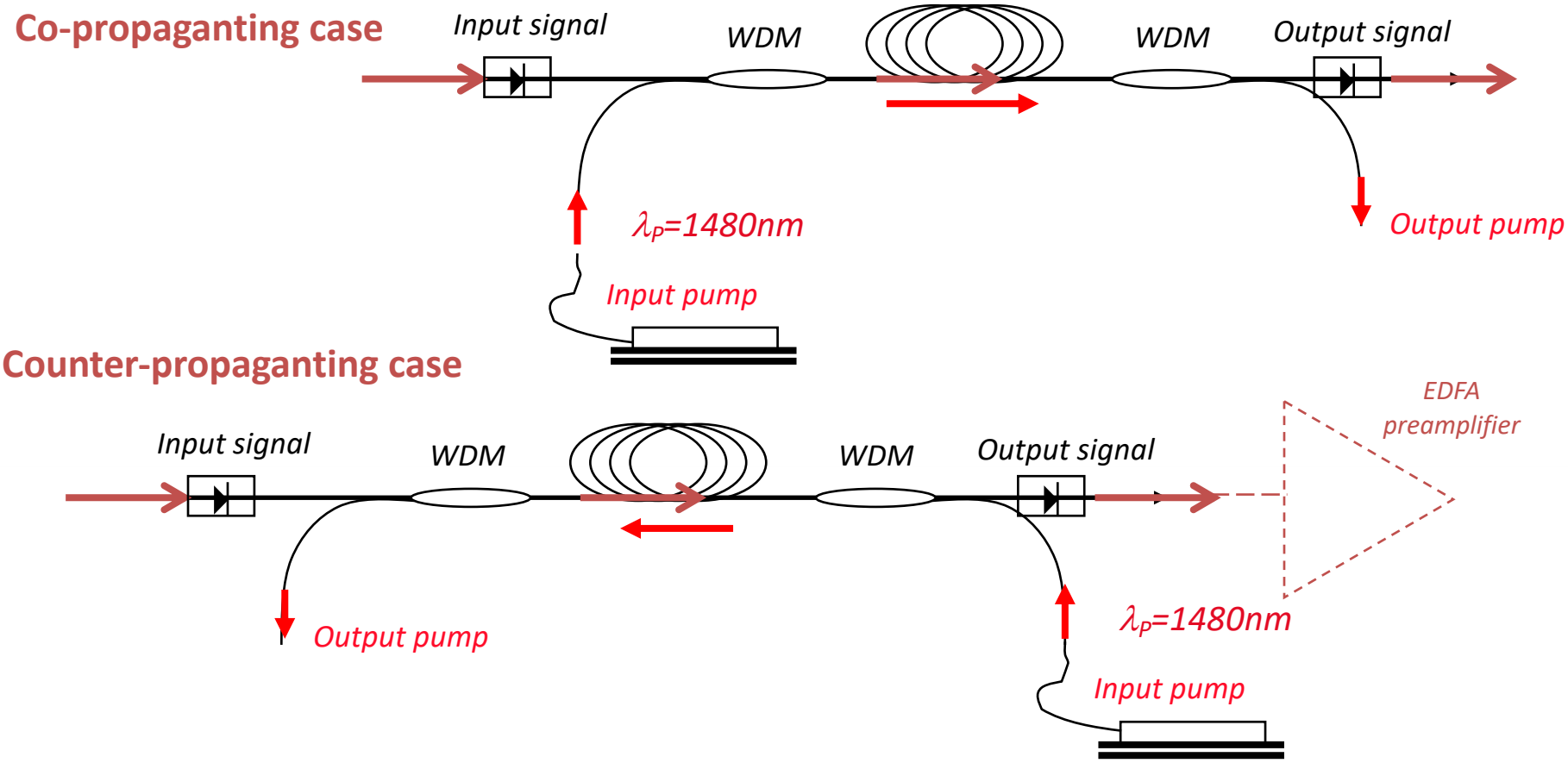


EDFA + Raman amplification bands



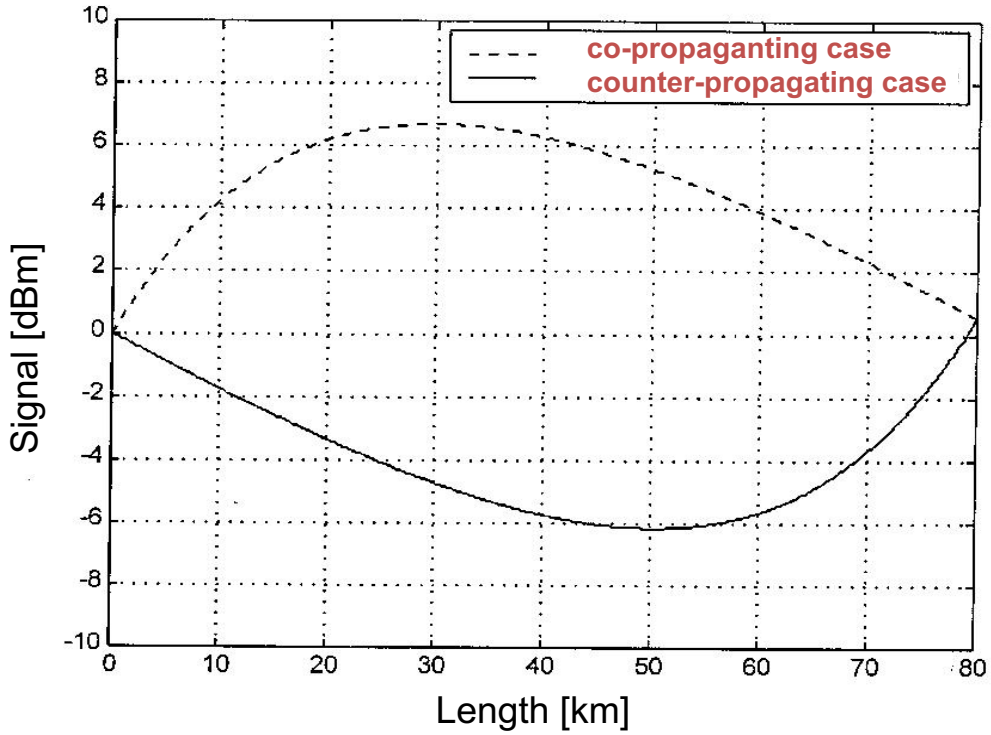
Raman distributed amplifier

- While the **EDFA is a lumped amplifier**, the Raman effect realizes a **distribued amlification.**
- Raman amplification is achieved both in case of pump and signal co-propagating in the fiber, and in case of pump and signal counter-propagating in the fiber.



Raman distributed amplifier

- **The counter-propagating configuration is preferable** because:
  - at the same power of the output signal, the signal level during propagation remains lower ==> **fewer non-linear effects**
  - possible **pump amplitude fluctuations are less transferred onto the signal.**



Advantages of Raman amplification

- Being distributed along the link, Raman amplification allows to reduce the signal power launched in input, **reducing non-linear effects** (SPM, XPM, FWM).
- Amplification **intrinsically less noisy** than the insertion of EDFA line amplifiers (the pump still makes spontaneous Raman scattering, subsequently amplified by SRS).
- **Spectral flexibility** of the gain band (also S band and L band).